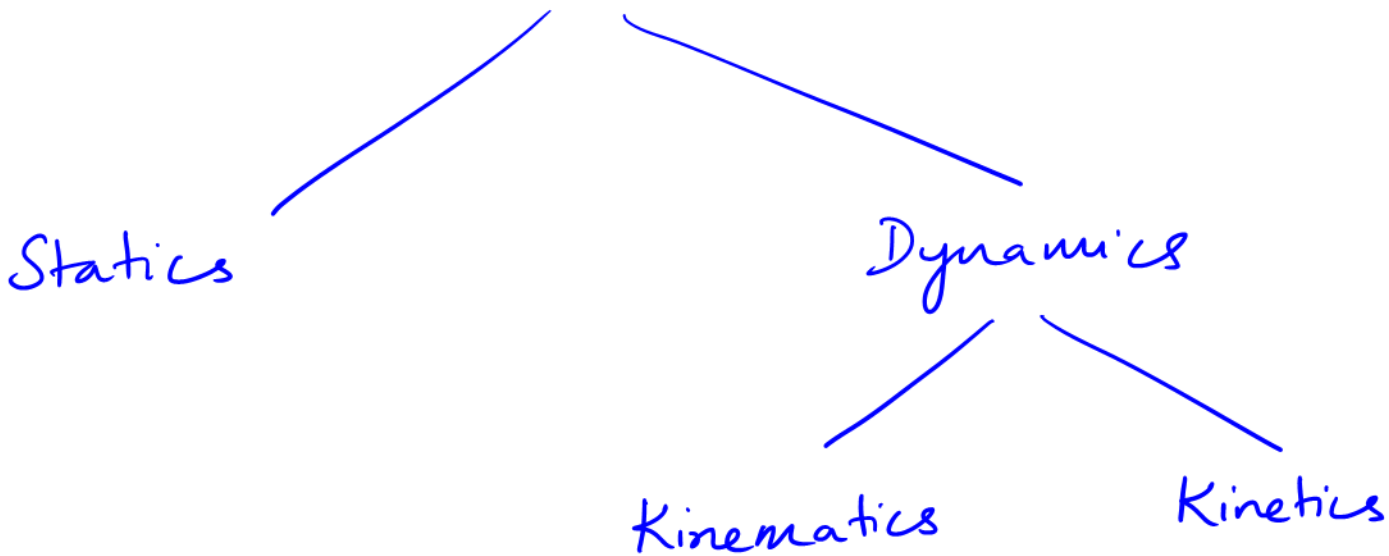
Mechanics of Human Movement

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Biomechanics - uses physics principles to quantitatively study how forces interact with the human body

Focus : Rigid body mechanics



Biomechanics : "body" could refer to the whole body or any of its parts or segments under consideration

Movements and locations are expressed w.r.t to reference anatomical planes, which are defined w.r.t the Anatomical Position

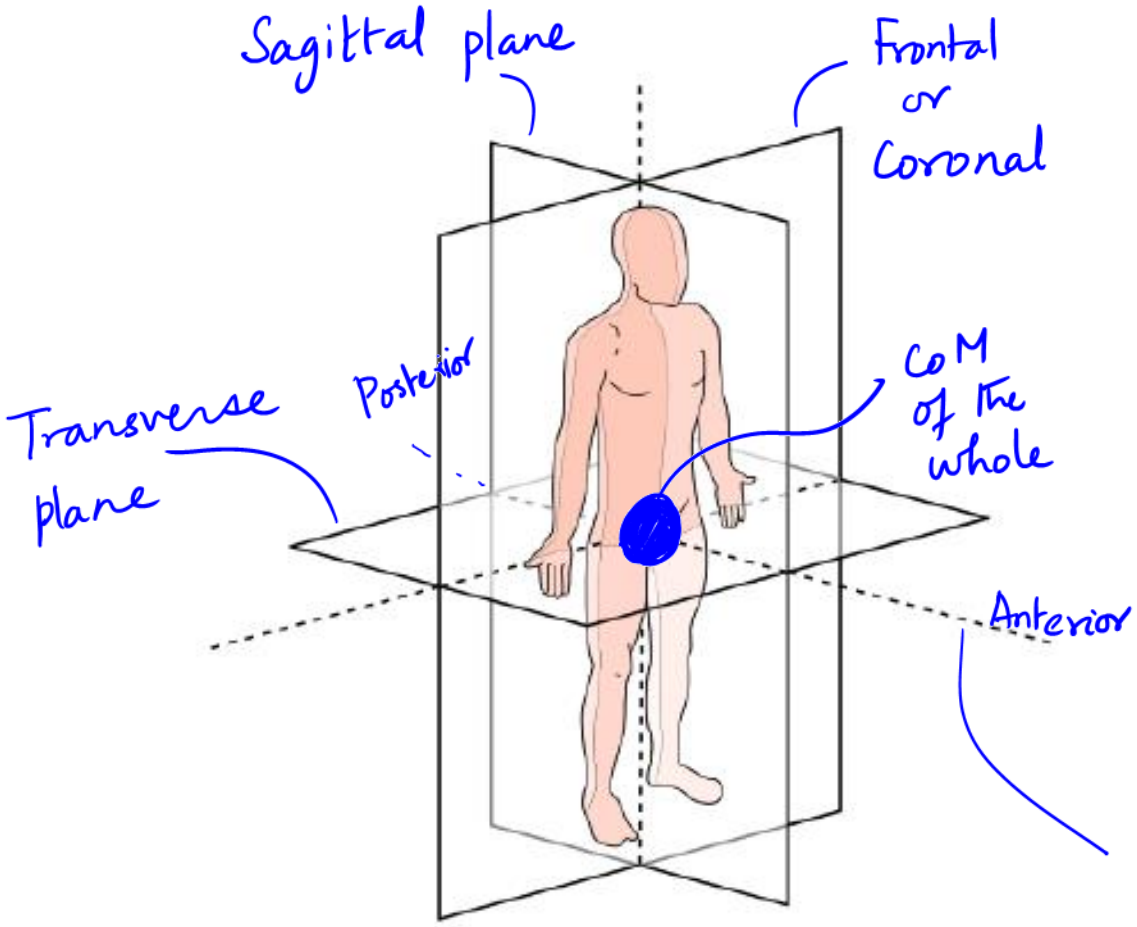
Musculoskeletal system

- Bones – structural members, assumed to be rigid
- Muscles - actuators
- Ligaments – limit motion
- Tendons - springs

connect bone to bone

connect muscle to bone

Anatomical position and planes



Anatomical posn: palms facing forward

3 principal or cardinal planes

Sagittal plane – bisects the body into left & right parts

Frontal plane – divides the body into front & back halves

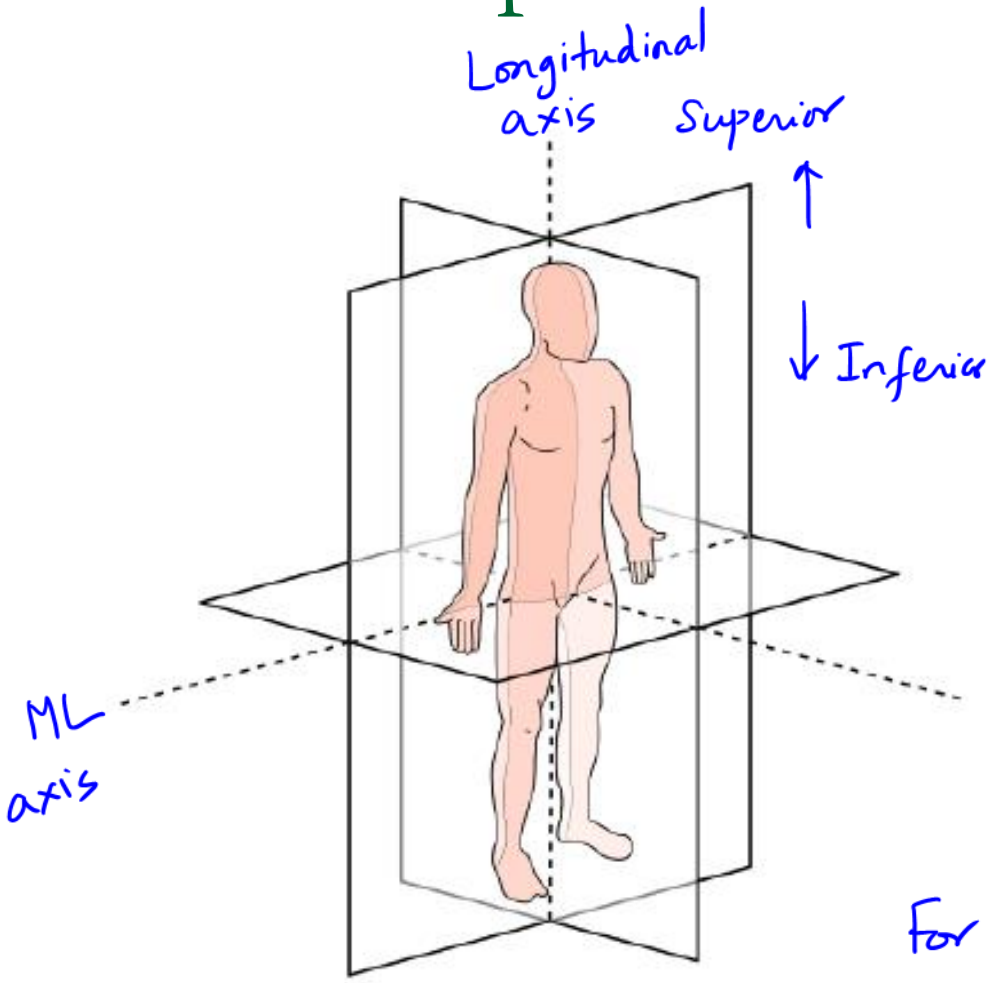
Transverse plane – divides the body into top & bottom halves

Axis ⊥r to Frontal plane – Anterio Posterior AP axis

Axis ⊥r to Sagittal plane – Mediolateral axis

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Anatomical position and planes



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Anatomical posn: palms facing forward

3 principal or cardinal planes

Medial – closer to the midline of the body

Lateral – farther from the midline of the body

Eye is medial to the ear

or Ear is lateral to the eye

Eyes are superior to the nose

or Nose is inferior to the eyes

For the limbs

Proximal – closer to its point of attachment to the body

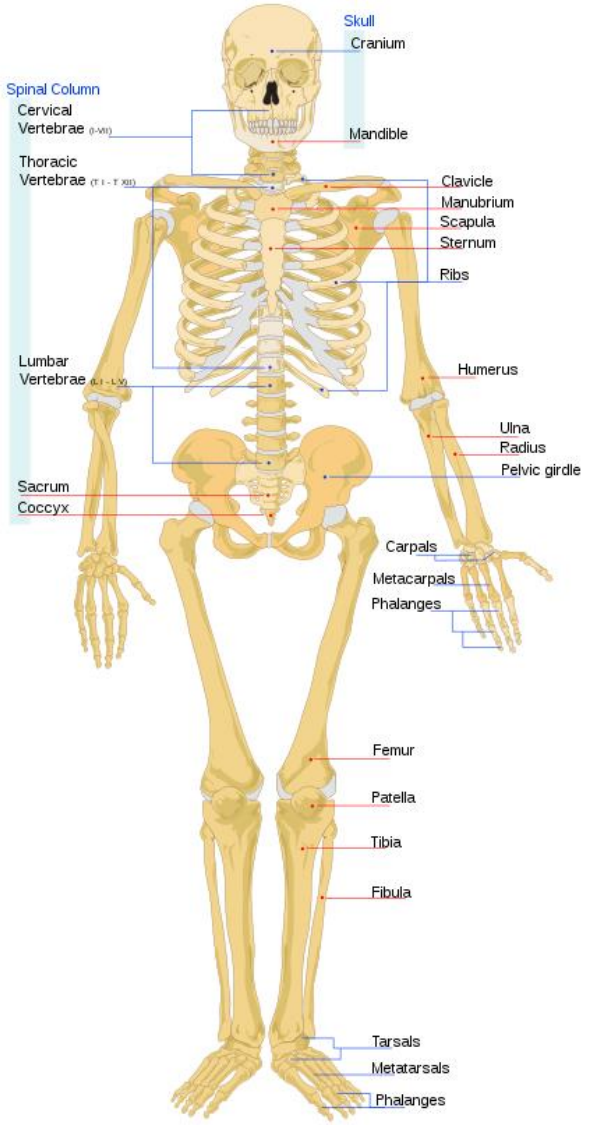
Distal – farther from the point of attachment

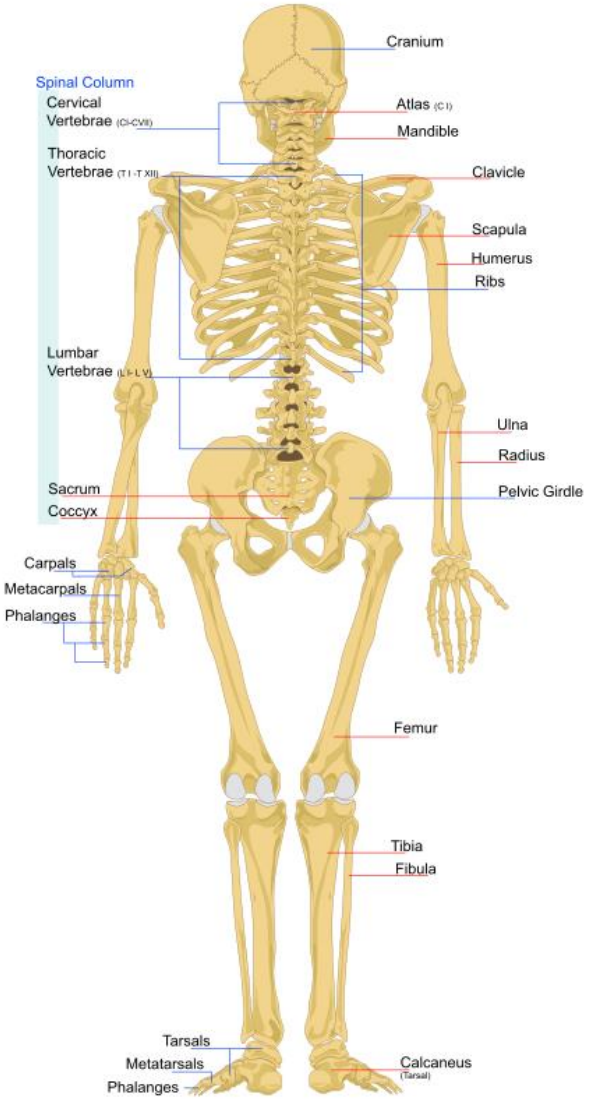
Skeletal system

Functions :

- Internal framework of body
- Supports the body
- Protects the internal organs
- Provides points of muscle attachment
- Produces blood cells
- Stores minerals

Bones Joints





206 bones in adults

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Bones are connected to each other to form the skeletal framework

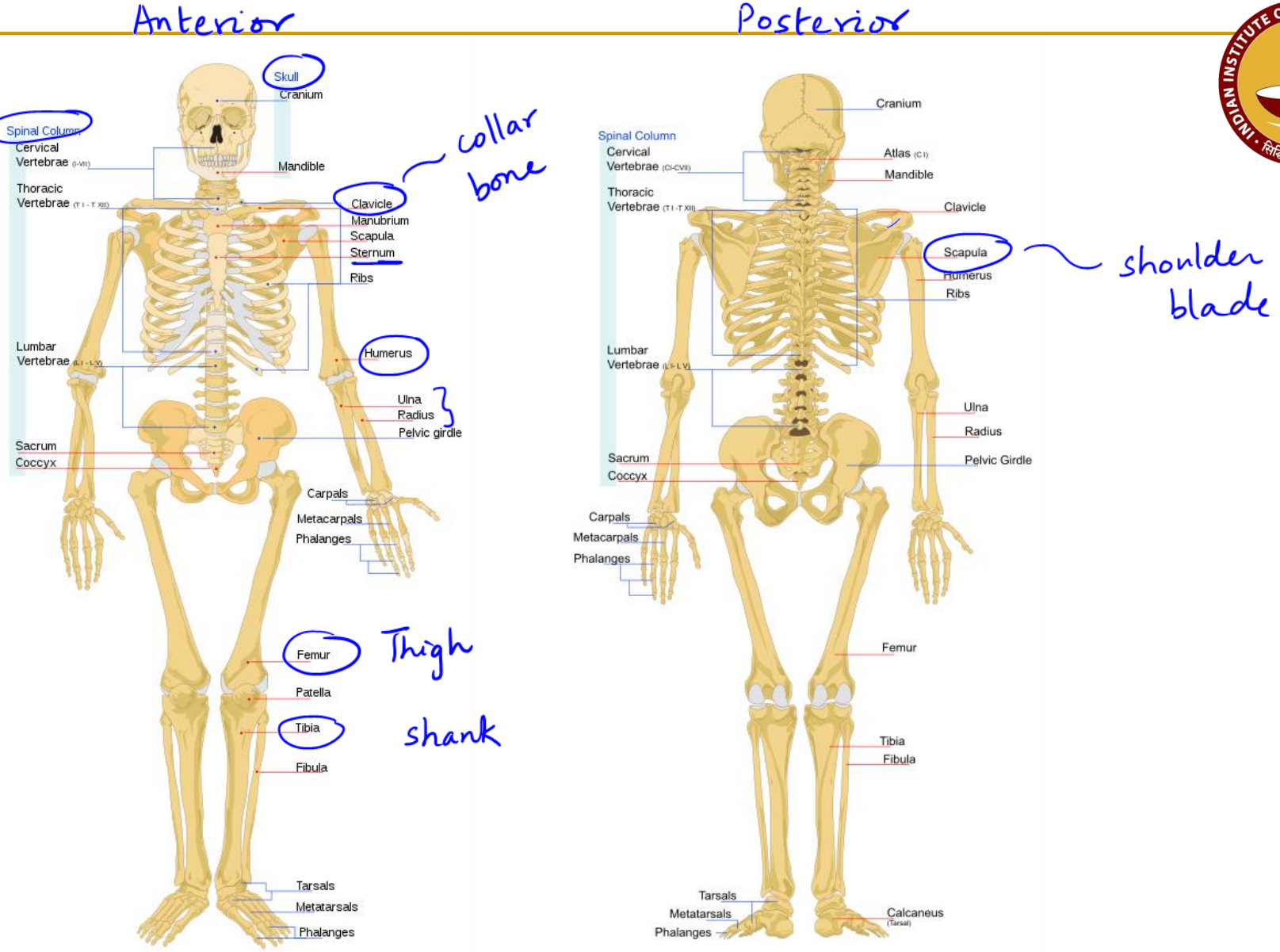
Joints – places where two bones meet
- are held together by ligaments
- give flexibility to the skeleton

Bones – also called osseous tissue, one of the hardest materials in the body

Cartilage → osteoblasts → osteocytes

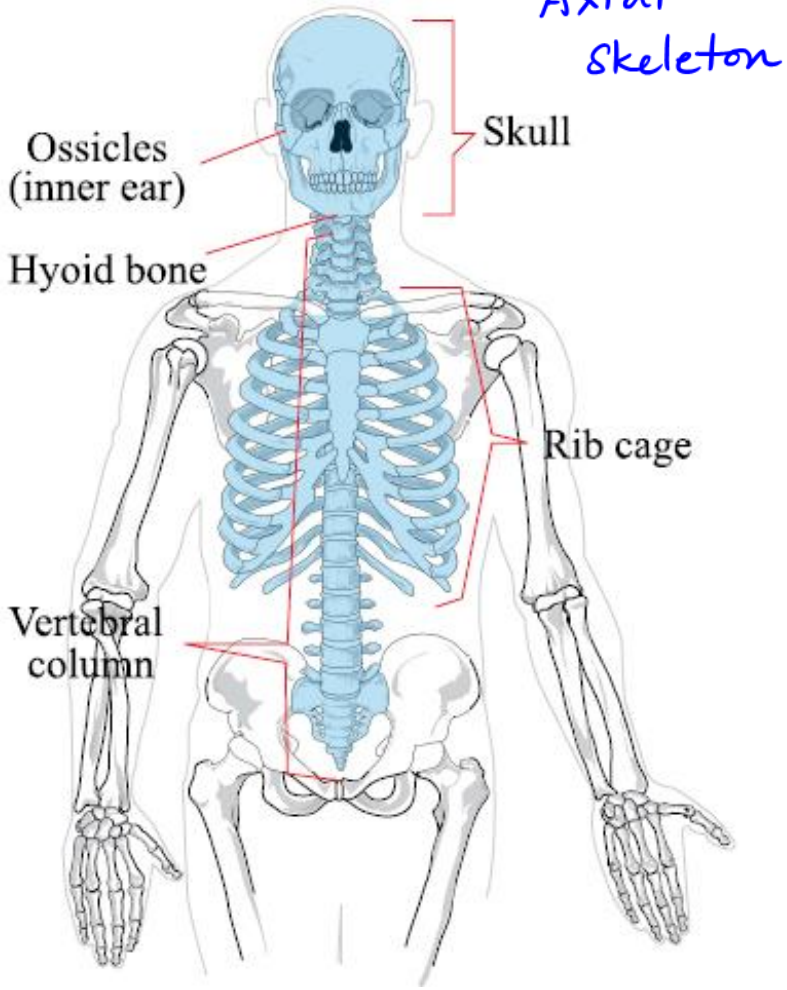
To form strong bones, need adequate minerals

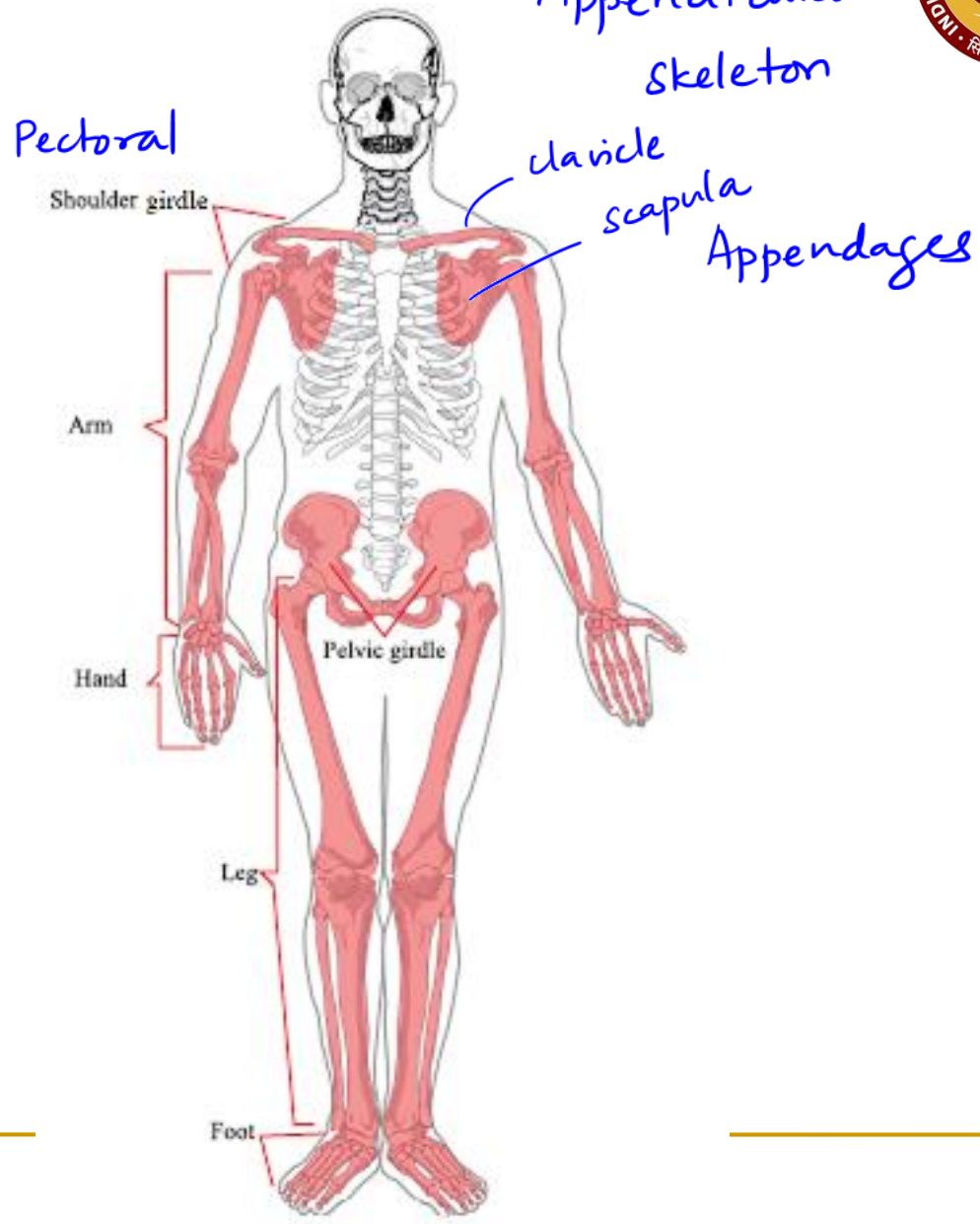
Skeletal system



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Skeletal system





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